Dario Serrano — Project Portfolio

Mechanical Engineering (Robotics & Mechatronics minor), University of Toronto — graduating June 2027
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Flight Control Engineering — HopTo Robotics

Flight Control Engineering Intern • Shenzhen & Hong Kong • Jun–Aug 2026

HopTo is the Shenzhen startup commercializing Prof. Pakpong Chirarattananon's hopping-robot research. I spent the summer on flight control and hardware for a novel water-hopping quadrotor that recovers energy from the water surface via a hydrofoil, alongside work on the company's production hopping robot. Full technical detail will appear with the IROS submission.

What I did

- Developed and tuned flight control for the water-hopping quadrotor; the work forms the basis of an IROS 2027 submission.
- Contributed mechanical design and bench testing to the hopping-robot platform, a 220 g machine carrying a 2 kg payload for inspection and survey work.
- Represented the company at the IEEE ICRA 2026 exhibition in Vienna, presenting hardware to researchers and industry attendees in English and Mandarin.

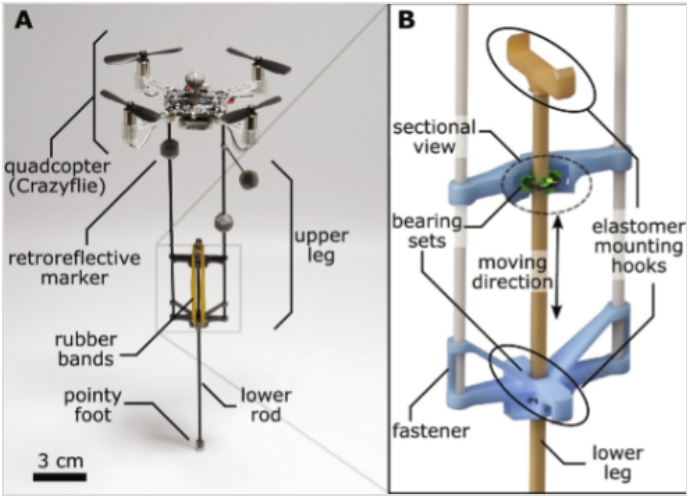
Why it mattered

- First sustained exposure to flight control on real hardware rather than in simulation, including the gap between the two.
- Led directly into continuing research at the CLIMB lab in Toronto and a first-author manuscript.






Hop-1
The first generation of Hopcopter


A
quadcopter
(Crazyflie)
retroreflective
marker
upper
leg
rubber
bands
pointy
foot
lower
rod
3 cm
B
sectional
view
bearing
sets
moving
direction
elastomer
mounting
hooks
fastener
lower
leg

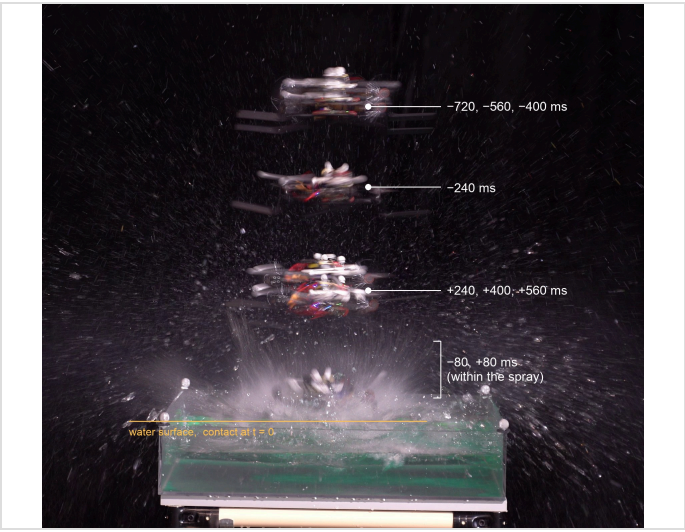
Revolving-Wing Quadrotor Flight Control — CLIMB Lab

Undergraduate Researcher • University of Toronto • Sep 2026 – present

Ongoing research in Prof. Chirarattananon's CLIMB lab on flight control for a revolving-wing quadrotor, as first author with a senior PhD collaborator. A manuscript is in preparation for IEEE Robotics and Automation Letters; full technical detail will appear with the paper.

What I am doing

- Leading flight control development for the platform, extending work begun during the summer in Shenzhen.
- Running experiments in the lab's motion-capture facility to characterize flight performance against millimetre-accurate ground truth.
- Preparing a first-author manuscript targeting IEEE Robotics and Automation Letters.



Ice-Perching Micro Air Vehicle — MIE491 Capstone

Team Lead • Client: Synetic Labs • Supervisor: Prof. Pakpong Chirarattananon • Sep 2026–Apr 2027

A hovering drone drains its battery in minutes; a perched one shuts down its motors and becomes a fixed sensor for hours or days. Published work has demonstrated perching on steep ice using spiny feet and dissipative landing gear, but only on platforms around 2 kg. I am leading a three-person team asking whether that architecture survives being scaled down to micro air vehicle size — and designing what replaces it where it does not.

What we are building

- A scaling analysis identifying where the existing architecture breaks at MAV scale. Impact energy falls by an order of magnitude, but landing-gear mass fraction becomes the binding constraint: spines and dampers have minimum viable sizes that a 150 g airframe cannot carry.
- A monolithic compliant mechanism — a single printed structure that dissipates impact energy, drives spine engagement and holds, replacing discrete springs, dampers and linkages.
- A drop-test rig characterizing maximum slope held, touchdown speed and holding force, with a documented ice-preparation protocol — ice is not a consistent material, and repeatability is a real experimental problem.
- Validation against motion-capture ground truth in the CLIMB lab facility.

My role

- Team coordinator across a three-person group, an industry client and a faculty supervisor.
- In charge of flight control development and R&D of the novel perching mechanism: scaling analysis, compliant-mechanism design, and drop-test characterization.

Androgynous Spacecraft Docking Mechanism

Kinematics & Dynamics of Machines • Sep–Dec 2025

In a senior-level mechanics course I led the design of a rough-alignment docking interface suitable for small satellite applications. The mechanism needed to be androgynous, compact, and capable of transmitting significant axial forces without jamming. Deliverables included a fully parametric CAD model, MATLAB analysis code, and a detailed final report.

What I built

- Six-arm variable-angle mechanism parametrized in SolidWorks to allow slider travel of 0.40–0.45 m and adjust the docking cone half-angle.
- MATLAB scripts calculating kinematics and static equilibrium; solved a 6×6 system for joint reactions under a 6 kN axial docking load (≈1000 N per arm).
- Jacobian-based force mapping to determine Stewart-platform-style actuator loads and inform actuator selection.
- FEA validation and parameter sweeps to ensure robustness across the motion range.

Key results

- Designed to withstand a 6 kN axial load while maintaining alignment tolerances.
- Calculated required slider actuator force of ≈2.17 kN, guiding motor sizing.

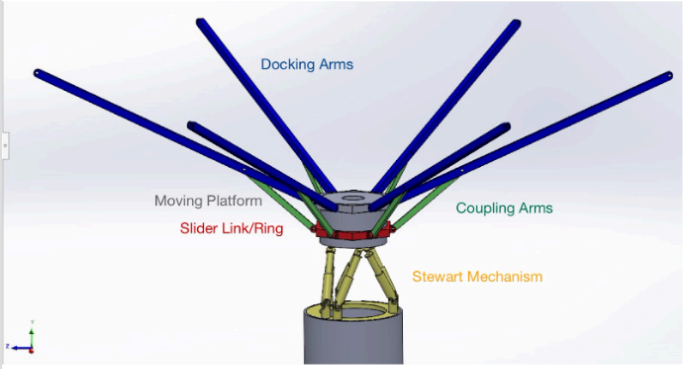


Figure 2. One half of proposed design outline and link labels

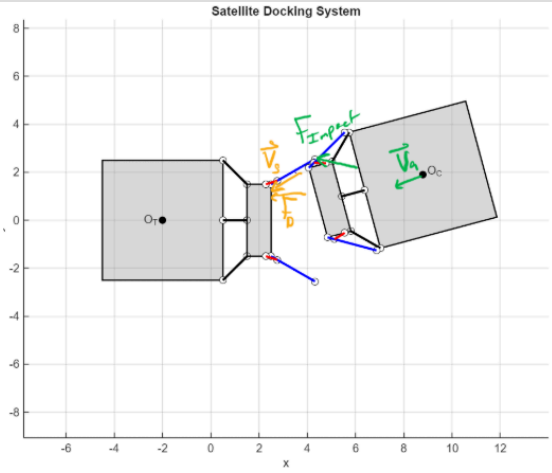


Figure 3. MatLab Docking Simulation

```
--- Geometry at docking configuration ---
A = (0.442, 0.000) m
B = (0.000, 0.000) m
C = (0.145, 0.039) m
D = (0.467, 0.125) m
Cone half-angle (B->D vs x-axis) = 15.00 deg

--- Forces for 6000 N total docking load (1000 N per arm) ---
Slider joint reaction Ax = 2165.4 N, Ay = -282.6 N
Required actuator force along slider (|Ax|) = 2165.4 N
Resultant force magnitude at slider joint = 2183.7 N
```

Fig.9 Force Component Calculations Printed in Command Terminal For α=15 degrees

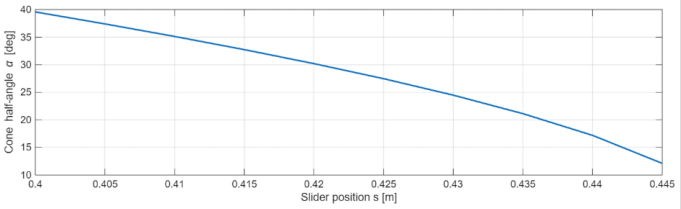


Fig.10 Cone half-angle vs. Slider Position for

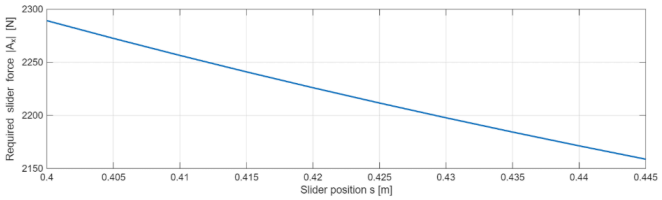


Fig.11 Required Slider Force vs. Slider Position

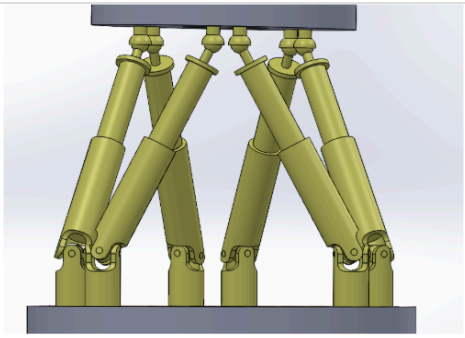


Fig.12 SolidWorks Model of Stewart Platform (Fine Alignment Mechanism)

EEG Seizure Detection with CNN–LSTM

Applied Fundamentals of Deep Learning • May–Sep 2025

This project tackled seizure classification on pediatric EEG recordings (CHB-MIT dataset). The goal was to build a deep learning pipeline that generalizes across patients by emphasizing robust preprocessing and evaluation on data the model had never seen.

What I built

- End-to-end PyTorch CNN–LSTM architecture trained on multi-channel EEG windows.
- Signal preprocessing including a 4th-order Butterworth bandpass filter, wavelet denoising, and overlapping sliding-window segmentation to boost data volume.
- Training/validation split arranged by patient to measure true generalization; evaluation metrics computed on held-out subjects.

Results

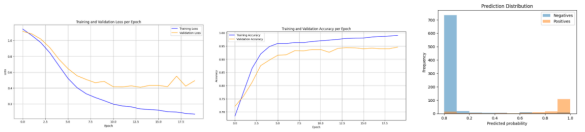
- AUROC: **0.96** demonstrating strong discrimination ability.
- F1-score: **0.84** on unseen patients, indicating balanced precision/recall.

Model	Accuracy	AUROC	F1	Recall	FPR
1D CNN+LSTM	0.944 ± 0.001	0.961 ± 0.002	0.835 ± 0.002	0.855 ± 0.005	0.135 ± 0.005
1D CNN+BLSTM	0.931 ± 0.004	0.947 ± 0.006	0.797 ± 0.009	0.80 ± 0.02	0.21 ± 0.02
2D CNN+LSTM	0.942 ± 0.003	0.964 ± 0.003	0.827 ± 0.009	0.803 ± 0.02	0.197 ± 0.02
2D CNN+BLSTM	0.933 ± 0.001	0.962 ± 0.003	0.844 ± 0.004	0.894 ± 0.008	0.146 ± 0.008

Table 1: Validation results of models averaged over three training sessions

7.2 QUALITATIVE RESULTS

To further illustrate how the CNN1DLSTM architecture performs, we present three representative plots generated from the training process.



(a) Training and Validation Loss over Epochs (b) Training and Validation Accuracy over Epochs (c) Prediction Probability Distribution

Figure 4: Training and prediction graphs of our final model.

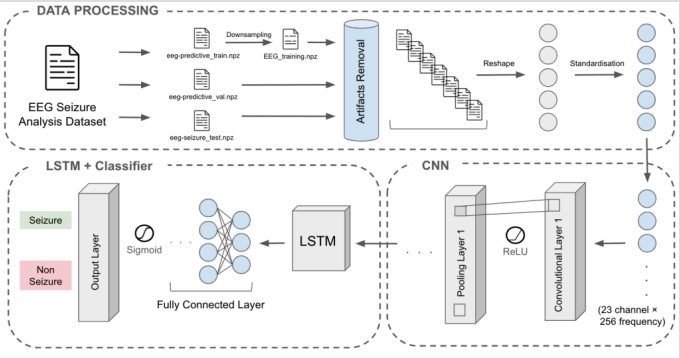


Figure 3: Neural Network Architecture

```
Final Test Classification Report:
              precision    recall  f1-score   support

         0.0     0.9653    0.9738    0.9695       801
         1.0     0.8671    0.8303    0.8483       165

    accuracy                         0.9493       966
   macro avg     0.9162    0.9020    0.9089       966
weighted avg     0.9486    0.9493    0.9488       966
```

Figure 5: Model Performance on Test Set

Current Sensing & Anti-Aliasing PCB

Digital and Analog Circuits • Jan–May 2026

Designed and fabricated a two-layer printed circuit board to interface current sensing with high-speed ADC sampling, incorporating analog filtering and anti-aliasing to support 50–70 kHz measurements.

- Analog front-end using LM324 to map 0–3.5 A current to 0.25–3.3 V ADC range under a 0–19 V single-supply constraint.
- Third-order Chebyshev low-pass filter (1 kHz cutoff, ≥ 35 dB attenuation at 5 kHz) designed in LTspice; validated DC gain and frequency response.
- Layout in Eagle with careful power/ground return routing; generated Gerber/drill files and hand-soldered prototype for validation.

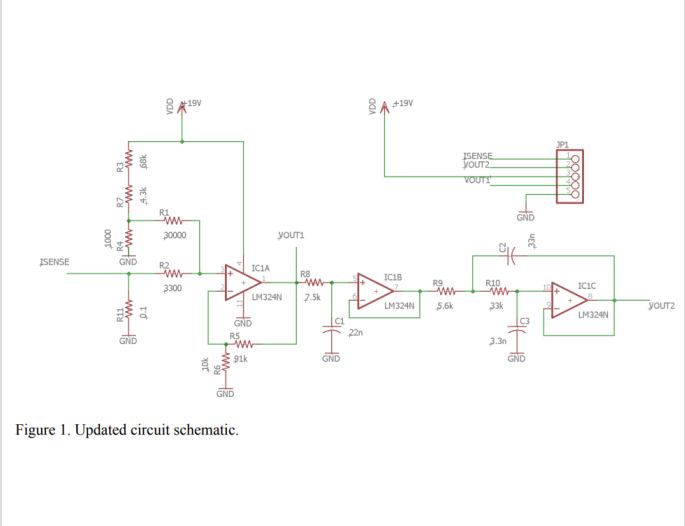
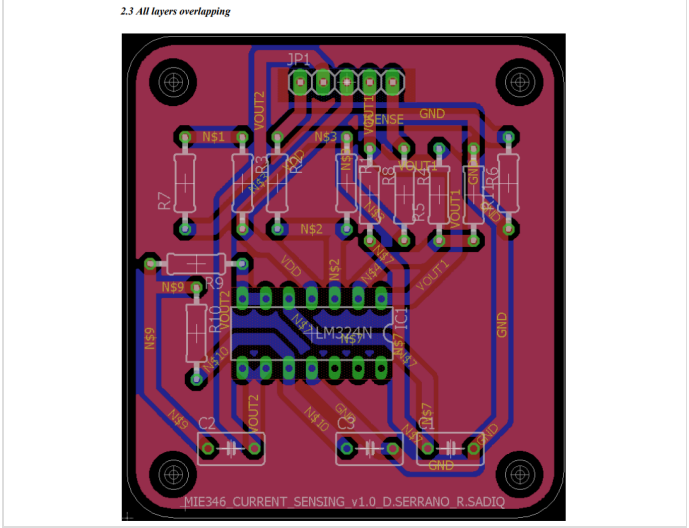


Figure 1. Updated circuit schematic.



CNC Mill Design Project

Mechanical Engineering Design • Sep–Dec 2024

Designed a desktop CNC mill targeting hobbyists, balancing cost, rigidity, and ease of manufacture. The project delivered full 3D CAD models, component selection rationale, and a detailed cost–performance analysis.

- Developed detailed 3D models of frame, linear axes, spindle assembly, and transmission in SolidWorks.
- Performed trade studies on drive options (stepper vs servo), linear guide types, and material choices to optimise accuracy per dollar.
- Documented fabrication processes, estimated BOM cost, and identified manufacturing challenges for a market-ready design.

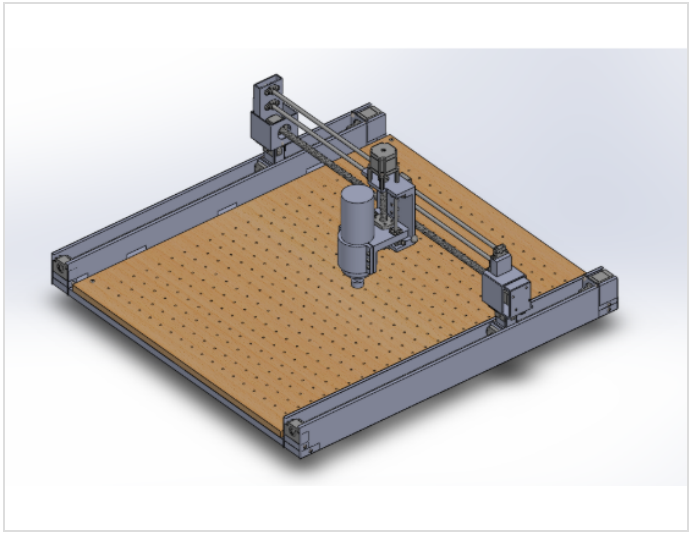
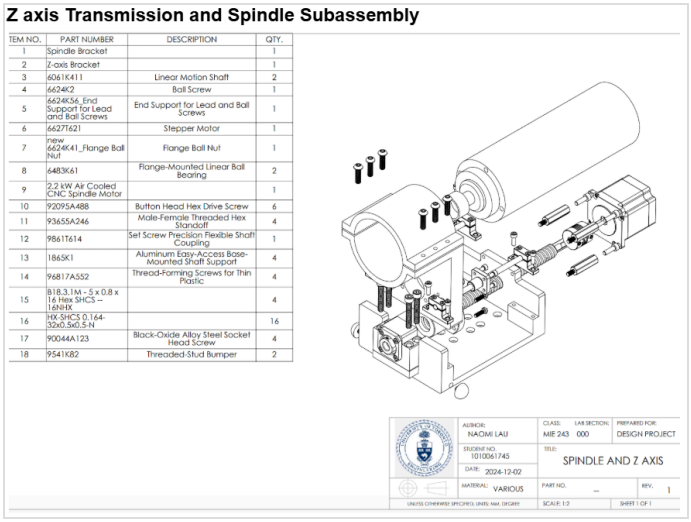
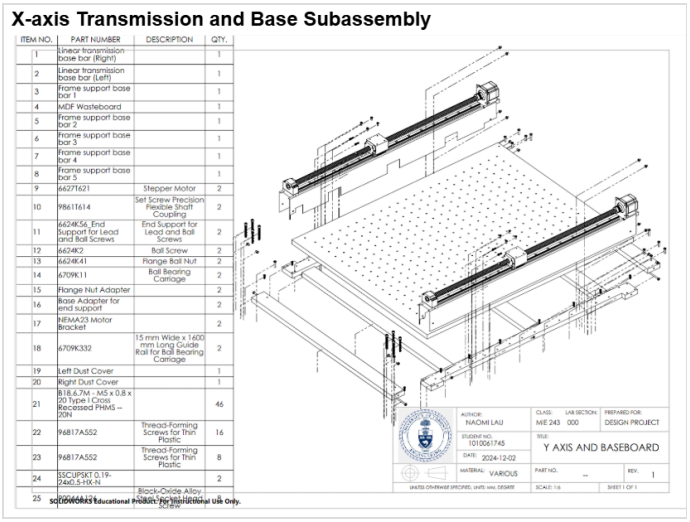


Table 12: X-Axis Components

Part Name	Quantity	Price/Unit [CAD]	Function	Source for CAD
Nema 23 Stepper Motor Bipolar 1.8 Deg 1.9Nm [34] 	2	$38.03	Used to power rotational motion to the ball screw, enabling linear transmission	Adapted from McMaster-Carr, Nema 23 Stepper Motor [35]
6.35mm Flexible Shaft Coupling [36] 	2	$3.62	Used for misalignment between ball screw and motor. Couples motor and ball screw shaft together	Adapted from McMaster-Carr, Set Screw Precision Flexible Shaft Coupling, [37]
SFU1605 Ball Screw and Flange Nut, 1000mm [38] 	2	$45.29 (Included in SFU1605 Ball Screw Kit)	Converts rotational motion from motor to linear motion to enable x axis travel	Adapted from McMaster-Carr, M16 Ball Screw, 1000mm [39] and M16 Flange Ball Nut [40]
BK12 End Support for Ball Screw [38] 	2	Included in SFU1605 Ball Screw Kit	Supports the end of the ball screw Resists axial and thrust forces in ball screw	Adapted from McMaster-Carr, End Support for Lead/Ball Screws [41]





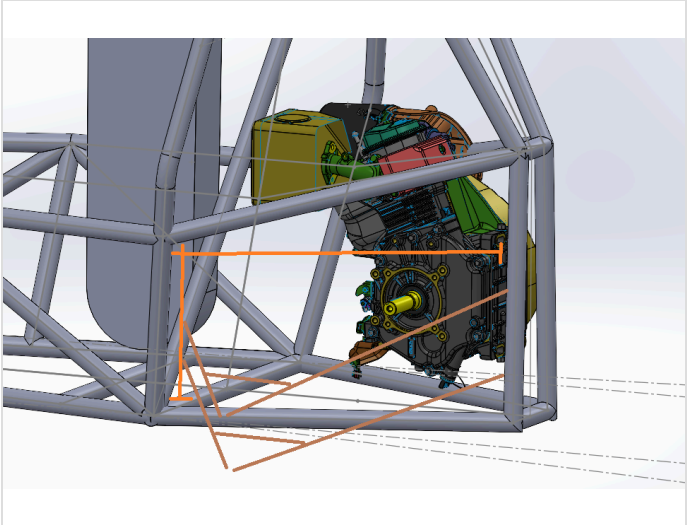
University of Toronto Baja Racing — SAE Baja

Suspension Team • Sep 2025–Present

Contributing member of a multidisciplinary engineering team re-establishing the University of Toronto Baja SAE racing program after six years of inactivity due to COVID-19. Working within the suspension sub-team, I helped build the team's first vehicle in six years, completed in 2026. The team is registered for the 2027 competition.

What I do

- Designed the front suspension A-arms and the brake hub to steering knuckle interface in SolidWorks.
- Perform hands-on fabrication including MIG/TIG welding and manual machining of suspension and chassis components to meet competition-ready tolerances.
- Develop original CAD models in SolidWorks for custom suspension components and tubular frame assemblies, ensuring compatibility with SAE rule-set constraints.
- Design around existing geometry from purchased off-the-shelf parts and components sourced from other university teams, integrating third-party hardware into the vehicle architecture.
- Collaborate with cross-functional sub-teams (drivetrain, steering, frame) to maintain system-level packaging and interface alignment across all vehicle subsystems.





Engineering Internship — Schindler Group

Shanghai • Jul–Aug 2024

During a summer internship with Schindler's global R&D team I assisted in structural analysis and prototyping for elevator components in an international setting.

- Parametric Creo modeling of Ω-Beam and L-Beam structural parts in collaboration with the worldwide R&D group.
- Finite element analysis performed according to FKM guidelines to verify strength under service loads.
- Developed MATLAB ODE models to analyze system oscillations and compare with simulation results.
- Produced 3D-printed prototypes for quick design iteration and evaluation on industrial printers.

Publication — Rocket Optimization

Jun–Oct 2022

“Applications of Optimization Techniques for Solid Rocket Design” (Darcy & Roy Press).
DOI: 10.54097/hset.v38i.5936 — https://doi.org/10.54097/hset.v38i.5936